



- (d) Given that $\int_0^1 \left(\frac{1}{2}\right)^x dx = \frac{1}{2 \ln 2}$, use the value of N found in part (c) to find upper and lower bounds for $\ln 2$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- 7 (a) Show that an appropriate integrating factor for

$$\sqrt{x^2 + 16} \frac{dy}{dx} + y = x\sqrt{x^2 + 16}$$

is $\frac{1}{4}x + \frac{1}{4}\sqrt{x^2 + 16}$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



